If $y = \sin^{-1}x$ prove that
$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$$

Here,

$y = \sin^{-1}$

differentiating wrt x

$$y_1 = \frac{1}{\sqrt{1-x^2}}$$

$$(1-x^2)y_1^2 = 1$$

$\boxed{\text{Squared both sides and multiplied by } (\sqrt{1-x^2})^2}$

differentiating wrt x

$$(1-x^2)(2y_1y_2) + (-2x)y_1^2 = 0$$

Dividing by $2y_1$

$$(1-x^2)y_2 - xy_1 = 0$$

applying Leibnitz theorem

$$(1-x^2)y_{n+2} + \left(\frac{n}{1!}\right)y_{n+1}(-2x) + \frac{n}{2!}y_n(-2) + \frac{n}{1!}y_n(1) + y_{n+1}x$$

or $(1-x^2)y_{n+2} - 2nxy_{n+1} - n(n-1)y_n - (xy_{n+1} + ny_n) = 0$

~~or $(1-x^2)y_{n+2} - 2nxy_{n+1}$~~

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - (n^2 - n + n)y_n = 0$

or, $(1-x^2)y_{n+2} - (2n+1)xy_{n+1} - n^2y_n = 0$ //

Here.

$$y = b\sin(\log x) + a\cos(\log x)$$

differentiating wrt x

$$y_1 = \frac{-a\sin(\log x)}{x} + \frac{b}{x}\cancel{\sin}\overset{\cos}{}(\log x)$$

$$xy_1 = -a\sin(\log x) + b\cos(\log x)$$

differentiating again:

$$xy_2 + y_1 = -\frac{a}{x}\cos(\log x) - \frac{b}{x}\sin(\log x)$$

$$x^2y_2 + xy_1 = -\left(a\cos(\log x) + b\sin(\log x)\right)$$

$$x^2y_2 + xy_1 + y = 0$$

differentiating n times using leibnitz theorem

$$\cancel{x^2y_{n+2}}$$

$$x^2y_{n+2} + \frac{2xn}{1!}y_{n+1} + 2\frac{n}{2!}y_n(2) + xy_{n+1} + \frac{n}{1}y_n + y_n = 0$$

~~or~~ $\cancel{x^2y_{n+2} + 2xny_{n+1} + 2ny_n + xy_{n+1} + ny_n + y_n = 0}$

or, $x^2y_{n+2} + 2xny_{n+1} + n(n-1)y_n + xy_{n+1} + ny_n = \cancel{0} - y_n$

or $x^2y_{n+2} + (2n+1)xy_{n+1} + n^2y_n - \cancel{ny_n} + \cancel{ny_n} + y_n = 0$

$\therefore$ $x^2y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$ //

PROVED

Evaluate:

$$\int_1^2\int_0^x \frac{1}{y^2+x^2}\,dy\,dx$$

Sol

$$= \int_1^2 \left[\frac{1}{x}\tan^{-1}\left(\frac{y}{x}\right)\right]_0^x dx$$

$$= \int_1^2 \left[\frac{1}{x}\tan^{-1}\left(\frac{x}{x}\right) - \frac{1}{x}\tan^{-1}\left(\frac{0}{x}\right)\right] dx$$

$$= \int_1^2 \left(\frac{1}{x}\cdot\frac{\pi}{4} - \frac{1}{x}\times 0\right) dx$$

$$= \int_1^2 \frac{\pi}{4x}\,dx$$

$$= \frac{\pi}{4}\int_1^2 \frac{1}{x}\,dx$$

$$= \frac{\pi}{4}\left[\ln(x)\right]_1^2$$

$$= \frac{\pi}{4}\left(\ln(2) - \ln(1)\right)$$

$$= \frac{\pi}{4}\left(\ln(2)\right) \quad \boxed{\because \ln(1) = 0}$$

CamScanner

Evaluate

$$\int_1^{\log 8}\int_0^{\log y} e^{x+y}\, dx\, dy$$

Sol

$$= \int_1^{\log 8}\int_0^{\log y} e^y e^x\, dx\, dy$$

$$= \int_1^{\log 8} \cancel{\int_0^{\log y}} e^y\left[e^x\right]_0^{\log y} dy$$

$$= \int_1^{\log 8} e^y\left(e^{\log y} - e^0\right) dy$$

$$= \int_1^{\log 8} e^y(y-1)\, dy$$

$$= \int_1^{\log 8} ye^y - y\, dy$$

$$= \left(ye^y - e^y - e^y\right)_1^{\log 8}$$

$$= \left[e^y(y-2)_+ \cancel{\log 8}\right]_1^{\log 8}$$

$$= e^{\log 8}(\log 8 - 2) - \left[e^1[1-2]\right]$$

$$= \cancel{\text{lo}}\ 8(\log 8 - 2) + e$$

$$= 8\log 8 - 16 + e$$

Find the reduction formula for $\int \sin^n x\, dx$ then evaluate for $\int \sin^6 x\, dx$

$$\int \sin^n x\, dx$$

$$= \int \sin^{n-1} x . \sin x\, dx$$

Applying integration by parts

$$= -\sin^{n-1} x \cos x - \int (-\cos x)(n-1) \sin^{n-2} x \cos x\, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x \cos^2 x\, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x (1-\sin^2 x)\, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) \int \sin^{n-2} x\, dx - (n-1) \int \sin^n x\, dx$$

$$= -\sin^{n-1} x \cos x + (n-1) I_{n-2} - (n-1) I_n$$

$$I_n + (n-1) I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$$

$$n I_n = -\sin^{n-1} x \cos x + (n-1) I_{n-2}$$

$$I_n = \frac{-\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} I_{n-2}$$

For $n=6$

$$I_6 = \frac{-\sin^{6-1} x \cos x}{6} + \frac{6-1}{6} I_{6-2}$$

$$I_6 = \frac{-\sin^5 x \cos x}{6} + \frac{5}{6} I_4$$

Similarly

$$I_4 = \frac{-\sin^3 x \cos x}{4} + \frac{3}{4} I_2$$

$$I_2 = \frac{-\sin x \cos x}{2} + \frac{1}{2} I_0$$

$$I_0 = \int \sin^0 x\,dx = \int 1\,dx = x$$

Combining all:

$$I_6 = \frac{-1}{6}\sin^5 x \cos x + \frac{5}{6}\left[\frac{-\sin^3 x \cos x}{4} + \frac{3}{4}\left(\frac{-\sin x \cos x}{2} + \frac{1}{2}x\right)\right]$$

$$\therefore I_6 = \frac{-1}{6}\sin^5 x \cos x - \frac{5}{24}\sin^3 x \cos x - \frac{15}{48}\sin x \cos x + \frac{15}{48}x + C$$

Reduction of $\int \sec^n x\,dx$ and solve for $\int \sec^7 x\,dx$

$I_n = \int \sec^{n-2}x \cancel{dx} \sec^2 x\,dx$

$= \sec^{n-2}x \tan x - (n-2)\int \sec^{n-2}x \tan^2 x\,dx$

$= \sec^{n-2}x \tan x - (n-2)\int \sec^{n-2}x(\sec^2 x - 1)dx$

$= \sec^{n-2}x \tan x - (n-2)I_n + (n-2)I_{n-2}$

$I_n(n-1) = \sec^{n-2}x \tan x + (n-2)I_{n-2}$

$$I_n = \frac{\sec^{n-2}x \tan x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

For $\sec^7 x$

$$I_7 = \frac{\sec^5 x \tan x}{6} + \frac{5}{6} I_5$$

$$I_5 = \frac{\sec^3 x \tan x}{4} + \frac{3}{4} I_3$$

$$I_3 = \frac{\sec x \tan x}{2} + \frac{1}{2} I_1$$

$I_1 = \int \sec x\,dx = \log|\sec x + \tan x|$

Combining all:

$$I_7 = \left(\frac{1}{6}\right)\sec^5 x \tan x + \frac{5}{24}\sec^3 x \tan x + \frac{5}{16}\sec x \tan x + \frac{5}{16}(\ln|\sec x + \tan$$

Using Beta gamma function:

$$\int_0^a x^2 (a^2 - x^2)^{3/2} \, dx \quad \text{——— (i)}$$

let $x = a\sin\theta$

$dx = a\cos\theta \, d\theta$,

When $x = 0 \rightarrow \theta = 0$

when $x = a$ $\theta = \frac{\pi}{2}$

So, (i) becomes:

$$\int_0^{\frac{\pi}{2}} a^2\sin^2\theta \left(a^2 - a^2\sin^2\theta\right)^{3/2} a\cos\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} a^2\sin^2\theta (a^2\cos^2\theta)^{3/2} \, a\cos\theta \, d\theta$$

$$= a^6 \int_0^{\frac{\pi}{2}} \sin^2\theta \, \cos^4\theta \, d\theta$$

using $\int_0^{\pi/2} \sin^p\theta \cos^q\theta \, d\theta = \dfrac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$

$p=2$, $q=4$

$$a^6 \cdot \frac{\Gamma(3/2)\,\Gamma(5/2)}{2\Gamma(4)} = a^6 \cdot \frac{\left((1/2)\sqrt{\pi}\right)\left(\frac{3}{4}\sqrt{\pi}\right)}{2(6)}$$

$$= \frac{\pi a^6}{32}$$

(11) $\displaystyle\int_0^{\pi/2} \sin^4 x \cos^2 x \, dx$

Applying $\displaystyle\int_0^{\pi/2} \sin^p\theta \cos^q\theta \, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$

$P=4$ $q=2$

$$= \frac{\Gamma\left(\frac{4+1}{2}\right)\Gamma\left(\frac{2+1}{2}\right)}{2\Gamma\left(\frac{4+2+2}{2}\right)}$$

$$= \frac{\left(\frac{1}{2}\sqrt{\pi}\right)\left(\frac{3}{4}\sqrt{\pi}\right)}{2(6)}$$

$$= \frac{\pi}{32}$$

~~First~~ find 1st and second partial derivative for

$$f(x,y) = x^2 - y^2 + 3xy$$

First order derivative:

$$f_x = \frac{\partial}{\partial x}(x^2 - y^2 + 3xy) = 2x + 3y$$

$$f_y = \frac{\partial}{\partial x}(x^2 - y^2 + 3xy) = -2y + 3x$$

Second order derivative

$$f_{xx} = \frac{\partial}{\partial x}(2x + 3y) = 2$$

$$\cancel{f_{yy}}\ f_{yy} = \frac{\partial}{\partial x}(-2y + 3x) = -2$$

$$f_{xy} = \frac{\partial}{\partial y}(2x + 3y) = 3$$

$$f_{yx} = \frac{\partial}{\partial x}(-2y + 3x) = 3$$

Verify Euler's theorem for:

① $u = f(x,y) = ax^2 + 2hxy + by^2$

② $u = x^n \sin\left(\frac{y}{x}\right)$

Eulers theorem If u is homogenous function of degree 'n'

then → $x\frac{du}{dx} + y\frac{\partial u}{\partial y} = nu$.

For ①

The given ~~std~~ equation

$$u(x,y) = ax^2 + 2hxy + by^2$$

all terms have ~~total~~ degree 2

then

$$u(tx,ty) = a(tx)^2 + 2h(tx)(ty) + b(ty)^2$$
$$= at^2x^2 + 2ht^2xy + bt^2y^2$$
$$= t^2(ax^2 + 2hxy + by^2)$$
$$= t^2u$$

$\therefore$ $n = 2$

partial derivative

$$\frac{\partial u}{\partial x} = 2ax + 2hy$$

$$\frac{\partial u}{\partial y} = 2hx + 2by$$

substituting in Eulers theorem

$x(2ax+2hy)+y(2hx+2by)$

$= 2ax^2+2hxy+2hyx+2by^2$

$= 2ax^2+4hxy+2by^2$

$= 2(ax^2+2hxy+by^2)$

$= 2u$

$= nu$

proved

(11) $u = x^n \sin\left(\frac{y}{x}\right)$

$u(tx,ty) = (tx)^n \sin\left(\frac{ty}{tx}\right)$

$= t^n\left(x \sin\left(\frac{y}{x}\right)\right)$

$= t^n(u)$

$\therefore$ degree is 'n'

$$\frac{\partial u}{\partial x} = nx^{n-1}\sin\frac{y}{x} + x^n\cos\left(\frac{y}{x}\right)\left(\frac{-y}{x^2}\right)$$

$$= nx^{n-1}\sin\left(\frac{y}{x}\right) - yx^{n-2}\cos\left(\frac{y}{x}\right)$$

$$\frac{\partial u}{\partial y} = x^n\cos\left(\frac{y}{x}\right)\left(\frac{1}{x}\right)$$

$$= x^{n-1}\cos\left(\frac{y}{x}\right)$$

Substituting in Eulers theorem

$$x\left(nx^{n-1}\sin\left(\frac{y}{x}\right) - yx^{n-2}\cos\left(\frac{y}{x}\right)\right) + y\left(x^{n-1}\cos\left(\frac{y}{x}\right)\right)$$

$$= nx^n\sin\frac{y}{x} \cancel{- yx^{n-2}\cos\left(\frac{y}{x}\right)} \cancel{+ yx^{n-1}\cos\left(\frac{y}{x}\right)}$$

$$= nx^n\sin\left(\frac{y}{x}\right)$$

$$= nu$$

// proved.

If $u = \log(x^3+y^3+z^3-3xyz)$ then prove:

(i) $$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$$

(ii) $$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)}$$

$$\frac{\partial u}{\partial x} = \frac{3x^2-3yz}{x^3+y^3+z^3-3xyz} = \frac{3(x^2-yz)}{x^3+y^3+z^3-3yz}$$

Similarly:

$$\frac{\partial u}{\partial y} = \frac{3(y^2-xz)}{x^3+y^3+z^3-3xyz}$$

and

$$\frac{\partial u}{\partial z} = \frac{3(z^2-yz)}{x^3+y^3+z^3-3xyz}$$

Adding them

$$\frac{3(y^2+x^2+z^2-xy-yz-zx)}{x^3+y^3+z^3-3xyz}$$

or $$= \frac{3\,\cancel{(x^2+y^2+z^2-xy-yz-zx)}}{(x+y+z)\cancel{(x^2+y^2+z^2-xy-yz-zx)}}$$

$$= \frac{3}{x+y+z} \quad \text{proved}$$

(ii)

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u$$

$$= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)\left[\frac{\partial u}{\partial x} + \frac{du}{\partial y} + \frac{\partial u}{\partial z}\right]$$

Substituting from (i)

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$$

partials of $\frac{3}{x+y+z}$

$$\frac{\partial}{\partial x}\left(\frac{3}{x+y+z}\right) = \frac{-3}{(x+y+z)^2}$$

Similarly

$$\frac{\partial}{\partial y}\left(\frac{3}{x+y+z}\right) = \frac{-3}{(x+y+z)^2}, \quad \frac{\partial}{\partial z}\left(\frac{3}{x+y+z}\right) = \frac{-3}{(x+y+z)^2}$$

Adding them

$$\frac{-3-3-3}{(x+y+z)^2} = \frac{-9}{(x+y+z)^2} \quad \text{proved}$$